Watching over Earth

From their orbits high above Earth, dozens of satellites keep watch over our planet. They range from surveillance satellites photographing enemy territory to weather satellites providing accurate forecasts and remote-sensing satellites that use advanced technology to map Earth's geology, land use, and changing climate.

LANDSAT 1

Few people guessed how useful it would be to study Earth from space until early astronauts reported seeing tiny details with the unaided eye and space agencies investigated further. The first satellite designed specifically for "remote sensing" of Earth was NASA's Landsat 1. Launched in 1972, it used a camera system and a multispectral scanner to study Earth's landmasses, including their agriculture, forestry, mineral resources, and water.

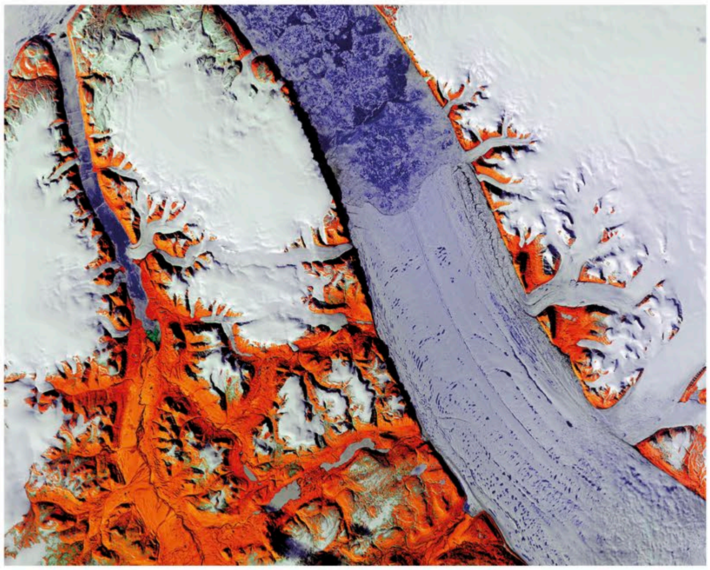
WOW!

In 1967, American spy satellites monitoring nuclear tests discovered blasts of gamma rays from galaxies billions of light-years away.

SPIES IN THE SKY

Spy satellites from the late 1950s onward paved the way for remote sensing. They used cameras with telephoto lenses to photograph enemy territory, returning reels of film in capsules that were parachuted back to Earth and snatched from midair by retrieval aircraft. When digital imaging improved to match the quality of film images, pictures could be sent by radio waves.

▼ MONITORING GLACIERS
False-color and infrared imaging highlights ice-free ground (red) around the Petermann Glacier (blue) in Greenland.



MULTISPECTRAL IMAGING

One of the most useful forms of remote sensing involves photographing the landscape through different colored filters, known as multispectral imaging. The brightness of the ground at different wavelengths can reveal information ranging from soil conditions and crop growth to the location of mineral resources and underground water.